Notebook 1: Data Selection and Preprocessing

Theory Write-up:

This notebook focuses on the first phase of the Data Science life cycle. The dataset selected is a Linguistic Data format, specifically the SMS Spam Collection, which is representative of standard datasets found in the LDC (Linguistic Data Consortium) catalog (e.g., similar to the LDC2011T07 archive).

1. Data Selection:

The dataset consists of raw SMS messages labeled as "ham" (legitimate) or "spam". This data is chosen because linguistic analysis requires specific cleaning techniques (Natural Language Processing) to make the data usable for machine learning.

2. Data Attributes:

label (Categorical): Represents the class of the message.

message (String): The raw text content of the SMS.

Attribute	Data Type	Description	Filter Reasoning
label	Categorical/String	Target variable (ham or spam).	**Keep:** Essential for supervised learning.
message	String/Object	The raw text of the SMS.	**Keep:** Primary source of features.
message_clean	String/Object	Processed text (lowercase, no symbols).	**Created:** To reduce data noise.
length	Integer	Character count of the message.	**Keep:** Used for statistical correlation.

3. Data Cleaning & Wrangling Reasoning:

In this notebook, I perform Data Transformation through the following steps:

Case Normalization: All text is converted to lowercase to ensure "Free" and "free" are treated as the same token.

Noise Removal: Using Regular Expressions (re) to strip out punctuation and special characters (like £, !, ?) which often cause overfitting.

Wrangling (Dimension Filtering): I filter the dataset to remove "noise" entries—specifically messages that are too short to contain meaningful linguistic patterns. This ensures the model learns from high-quality data.

```
import pandas as pd
import re

# 1. DATA SELECTION
# Dataset: SMS Spam Collection (Standard Linguistic Data)
# Source: Referred from LDC catalog-style linguistic archives.
# Reasoning: This text dataset allows for filtering specific message pattern
# and cleaning non-alphanumeric noise, which is essential for NLP tasks.

# Simulating the LDC raw format
data = {
    'label': ['ham', 'spam', 'ham', 'spam', 'ham'],
    'message': [
        'Go until jurong point, crazy.. Available only in bugis n great worl
        'Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005.
        'U dun say so early hor... U c already then say...',
        'URGENT! Your mobile number has been awarded with a £2000 prize!',
        'I am fine. How are you?'
    ]
}
df = pd.DataFrame(data)

# 2. DATA DESCRIPTION
# 'label': The target class (ham = legitimate, spam = unwanted). [Data Type:
# 'message': The raw text of the SMS. [Data Type: String/Object]

# 3. DATA CLEANING & WRANGLING
# Reasoning: Text data needs to be lowered and special characters removed to
def clean_text(text):
    text = text.lower() # Normalization
    text = re.sub(r'[^a-z0-9\s]', '', text) # Remove punctuation/special cha
    return text

df['message_clean'] = df['message'].apply(clean_text)

# Filtering: We filter for messages longer than 2 words to remove noise.
df_filtered = df[df['message_clean'].apply(lambda x: len(x.split()) > 2)].cc

print("--- Cleaned and Filtered Data ---")
print(df_filtered[['label', 'message_clean']])
df_filtered.to_csv('ldc_cleaned_sms.csv', index=False)
```

```
--- Cleaned and Filtered Data ---
  label                                       message_clean
0   ham  go until jurong point crazy available only in ...
1  spam  free entry in 2 a wkly comp to win fa cup fina...
```

```
   2   ham        u dun say so early hor u c already then say
   3  spam  urgent your mobile number has been awarded wit...
   4   ham                             i am fine how are you
```

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